Krishnaa **Vadivel** | Computational Physics PhD | First-Principles Materials | HPC | Scientific ML

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Summary

Computational physics researcher combining first-principles condensed-matter theory, many-body modeling, and production scientific software. Experience spans distributed HPC on Frontier and Perlmutter, CUDA/ROCm acceleration, semiconductor simulation, and ML methods for materials and molecular systems.

Research Experience

Yale University, Electrical Engineering
PhD Research Assistant *2021–present*

- Develop first-principles and many-body approaches for self-trapped excitons and excitonic polarons in real materials, connecting DFT, DFPT, GW/BSE-style workflows, and reduced effective models.
- Build distributed simulation workflows in Python and C++ for leadership-class HPC systems, using MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc on Frontier, Perlmutter, and other national lab clusters.
- Formulate coherent-state and path-integral viewpoints for exciton-phonon dynamics and investigate ML accelerators, including equivariant graph neural networks, for faster materials modeling.

Industry Experience

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*

- Developed CUDA and OpenGL software for high-volume acoustic waveform processing, visualization, and field-tool diagnostics in geological well logging systems.
- Implemented embedded and sensor-side functionality across Lattice FPGA, PIC32, dsPIC, STM32, and C8051 platforms; supported circuit bring-up, oscilloscope-based debugging, and selective PCB updates in Altium.
- Contributed host-side tooling with C#, ASP-based services, and Microsoft SQL-backed workflows for data handling and engineering operations.

FindLight
Photonics Technical Writer *2018–2019*

- Wrote clear technical content on lasers, optics, and photonics devices for engineering audiences and product-focused readers.

Selected Projects

Exciton-Phonon HPC Research: Distributed first-principles calculations for self-trapped excitons, many-body formulations, and scalable workflows for materials simulations.

Semiconductor Device Simulation: Numerical modeling of semiconductor equations, transport, and pn-junction-style device behavior for physics and EE audiences.

Equivariant Molecular ML: Graph neural network workflows for molecular-property prediction and ML-assisted scientific computing.

Agentic Scientific Automation: RAG-style tooling for reading documentation and research papers to generate structured computational-science inputs and workflows.

Education

Yale University
PhD, Electrical Engineering *2021–present*

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript
Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, SciPy, SymPy, matplotlib
ML: PyTorch, PyTorch Geometric, scikit-learn
Platforms: Linux, macOS, Windows, Docker, Docker Compose, Kubernetes, gcloud
Domains: First-principles materials modeling, many-body theory, semiconductor physics, numerical PDEs